

A Statistical Audit of LLM Calibration Heterogeneity Across Knowledge Domains

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STAT 215B, Spring 2026

Abstract

Large language model (LLM) calibration is nearly always reported as a single aggregate scalar, obscuring whether miscalibration is diffuse or concentrated in specific knowledge domains. We develop a statistically rigorous audit framework and apply it to MMLU [Hendrycks et al., 2021] using token-level log probabilities from four instruction-tuned models spanning two families and three parameter scales: GPT-4o-mini, Llama-3-8B-Instruct, Qwen-2.5-1.5B-Instruct, and Qwen-2.5-7B-Instruct. Our framework combines a three-level mixed-effects model, isotonic and kernel-smoothed reliability curves, empirical Bayes shrinkage, and Benjamini–Hochberg FDR correction. Across all four models, between-subject calibration variance accounts for 1.9–7.9% of the total, and 54–57 of 57 subjects are flagged as significantly miscalibrated at FDR $\alpha = 0.05$. Subject-level ECE rankings are highly correlated across all model pairs (Spearman ρ between 0.78 and 0.92), indicating that miscalibration is task-driven rather than model-specific. The same subjects are robustly worst-calibrated across the four models (college-level quantitative STEM, niche factual recall, and structured ethical/legal reasoning) and the same subjects are robustly best-calibrated (high-school-level content in Social Sciences and applied/factual subjects). Question-level entropy enters the multilevel model with a coefficient that varies in sign across the four checkpoints—positive for GPT-4o-mini and Qwen-7B, marginal for Llama, negative for Qwen-1.5B—a marginal pattern consistent with prior aggregate findings on RLHF-induced confidence saturation [Kadavath et al., 2022, Nakkiran et al., 2025].

1 Introduction

Large language models are increasingly deployed in high-stakes settings—healthcare, law, education—where the reliability of model-expressed confidence is as important as accuracy itself. A well-calibrated model is one whose stated confidence tracks its empirical accuracy: if a model assigns 80% probability to an answer, it should be correct 80% of the time. Miscalibration, particularly overconfidence, means that users who rely on model confidence as a decision signal will be systematically misled.

A substantial literature has studied LLM calibration. Guo et al. [2017] established the modern empirical study of neural network calibration, popularizing expected calibration error (ECE) as a standard metric and demonstrating the effectiveness of temperature scaling as a post-hoc correction. Kadavath et al. [2022] showed that for multiple-choice tasks, token-level log probabilities provide a meaningful calibration signal, and that base (pre-RLHF) models exhibit better-calibrated token probabilities than instruction-tuned counterparts. Xiong et al. [2024] provided a systematic benchmark of confidence elicitation methods, finding that all methods “struggle in challenging tasks, such as those requiring professional knowledge”—suggesting calibration quality may vary by domain, though the authors do not investigate this directly. Jiang et al. [2021] similarly found that

question-level features such as answer entropy predict calibration quality, a pattern we examine formally here.

What is missing is a *statistical audit* of that heterogeneity. To our knowledge, the existing literature reports subject-level ECE without (i) stabilizing estimates against large differences in per-subject sample size, (ii) formally testing which subjects are robustly miscalibrated after correcting for multiple comparisons, (iii) decomposing how much total calibration variance is attributable to domain versus subject versus individual question, or (iv) characterizing subjects by their *pattern* of miscalibration rather than its scalar magnitude.

This framing is analogous to the gap between per-group accuracy reporting and a rigorous fairness audit [Barocas et al., 2023]: the fairness literature long ago recognized that single-number summaries are insufficient for subgroup auditing, and we import that logic into calibration evaluation. The downstream implication of the audit is structural: if miscalibration is diffuse, a single global recalibration (e.g., temperature scaling) is a reasonable fix; if it is concentrated in specific subject clusters and those clusters are stable across models, then subject-aware recalibration becomes a structurally appropriate scoping choice—a position that has been validated empirically by Luo et al. [2025], who fit subject-specific temperature parameters per MMLU subject. Our contribution is the diagnostic arm of that picture: stable audit findings establish whether the structural assumption behind a subject-aware correction is warranted in the first place.

Our central scientific questions, restated:

1. **Where** is calibration good and bad? Variance decomposition, FDR-corrected subject rankings, and a within-MMLU contrast between well-calibrated and poorly-calibrated subject clusters.
2. **How consistent** is the answer across models? Pairwise Spearman correlations of subject-level ECE across four models.
3. **What mechanisms** produce the observed heterogeneity? Question- and subject-level predictors plus a within-family scale comparison.

2 Related Work

Calibration of neural networks. Guo et al. [2017] showed that modern neural networks are systematically overconfident and that temperature scaling largely corrects this at the aggregate level, with Zadrozny and Elkan [2002] earlier establishing isotonic regression as an alternative nonparametric recalibration method. Kadavath et al. [2022] extended calibration evaluation to LLMs, establishing that token-level probabilities on multiple-choice tasks carry a meaningful calibration signal. Nakkiran et al. [2025] and Yaldiz et al. [2026] study how instruction tuning affects calibration, finding that RLHF-style fine-tuning tends to degrade calibration relative to base models, an effect also documented in Ouyang et al. [2022].

Domain-level heterogeneity. Xiong et al. [2024] is the closest prior work in spirit, benchmarking confidence elicitation across task types. Jiang et al. [2021] showed that question-level features predict calibration on QA tasks. Luo et al. [2025] fits subject-specific temperature parameters for each of MMLU’s 57 subjects, implicitly acknowledging that a global correction is insufficient; Tan et al. [2026] uses base model signals for unsupervised recalibration. Neither work performs a formal statistical audit of subject-level heterogeneity.

Multiple testing and empirical Bayes. The Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995] controls the false discovery rate and is standard in high-dimensional inference. Empirical Bayes shrinkage of group-level estimates [e.g., Efron, 2012] is standard in multilevel modeling via best linear unbiased predictors (BLUPs). We use the multilevel modeling framework of Raudenbush and Bryk [2002] to decompose calibration variance across levels of the MMLU hierarchy.

3 Data

MMLU. We use the Massive Multitask Language Understanding benchmark [Hendrycks et al., 2021], accessed via the `cais/mmlu` dataset on HuggingFace. MMLU comprises 14,042 four-option multiple-choice questions across 57 subjects nested in four broad domains: STEM, Social Sciences, Humanities, and Other. Subject sizes range from 100 to 1,534 questions (median 163). The three-level hierarchy—questions $\rightarrow$ subjects $\rightarrow$ domains—is natural for the proposed analysis. We use the test split throughout.

Models. We query four instruction-tuned models spanning two model families and three parameter scales. **GPT-4o-mini** is queried via the OpenAI Batch API at temperature 0 (`max_tokens=1`, `top_logprobs=20`); the choice of 20 (rather than the API default of 5) ensures all four answer tokens are captured for virtually every question. **Llama-3-8B-Instruct** and **Qwen-2.5-{1.5B, 7B}-Instruct** are run locally on a Colab T4 with 4-bit quantization (BitsAndBytes; Dettmers et al. 2022). For all four instruction-tuned models we pre-fill the assistant turn with “The answer is” before extracting logprobs at the next-token position, placing the scoring position immediately before the answer token. Both pipelines compute the same quantity: the renormalized softmax probability over $\{A, B, C, D\}$. Test-set accuracies are 0.749 (GPT-4o-mini), 0.606 (Llama), 0.536 (Qwen-1.5B), and 0.685 (Qwen-7B); the open-weight models are slightly below their published full-precision figures, an effect we attribute to 4-bit quantization.

4 Methods

4.1 Confidence Elicitation

For each question, we extract the token-level log probabilities over the four answer tokens (A, B, C, D). Applying softmax renormalization yields a proper probability distribution over options; the model’s confidence $c_i \in (0, 1]$ is the maximum of this renormalized distribution, equal to $P(\text{predicted answer} \mid \text{answer} \in \{A, B, C, D\})$. This is the standard approach in calibration research [Kadavath et al., 2022] and avoids the noise of verbal elicitation.

For GPT-4o-mini, logprobs are drawn directly from the API’s top-20 response. For the open-weight models, logprobs are extracted from the full vocabulary softmax at the assistant-generation position after the “The answer is” prefix. All four pipelines compute the same quantity: $P(\hat{y} \mid \hat{y} \in \{A, B, C, D\})$.

4.2 Outcome and Covariates

Primary outcome. The primary outcome is the *signed calibration gap* at the question level:

$$\text{gap}_i = c_i - y_i, \tag{1}$$

where $y_i \in \{0, 1\}$ indicates correctness. This quantity is positive under overconfidence and negative under underconfidence, and supports regression-based inference that binning-based ECE cannot. Its mean at the subject level directly measures the direction and magnitude of miscalibration.

Question-level covariates. We extract the following features from each question:

- **Word count:** number of words in the question stem.
- **Max option length:** word count of the longest answer choice.
- **Negation indicator:** binary, flagging presence of *not*, *except*, *least*, *never*, or *no*.
- **Entropy:** Shannon entropy in nats of the softmax distribution over A/B/C/D, $H_i = -\sum_k p_{ik} \log p_{ik}$. Zero when one option dominates, $\log 4 \approx 1.386$ when uniform.

All continuous covariates are standardized before model fitting.

Heteroskedasticity note. The variance of $\text{gap}_i = c_i - y_i$ is not constant across subjects: since $y_i \sim \text{Bernoulli}(p_j)$, its contribution to variance is $p_j(1 - p_j)$, which is highest for subjects near 50% accuracy. We check residual plots for evidence of heteroskedasticity and note this as a potential limitation of the homoskedastic mixed model.

4.3 Nonparametric Calibration Curves

For each subject, we estimate a reliability curve using isotonic regression—a nonparametric monotone fit of empirical accuracy on confidence [Zadrozny and Elkan, 2002]—rather than arbitrary equal-width binning. The monotonicity constraint is theoretically motivated but is an assumption rather than a guarantee. We therefore fit an unconstrained kernel smoother (Gaussian, $\sigma = 1.5$ bins) alongside each isotonic curve; substantive divergences between the two identify subjects where the monotonicity assumption may distort the estimated curve.

ECE per subject is computed as the weighted mean absolute deviation between the reliability curve and the diagonal, using 20 equal-width confidence bins.

4.4 Multilevel Regression and Variance Decomposition

We model the question-level calibration gap across all 14,042 questions using a three-level mixed model with questions nested in subjects nested in domains:

$$\text{gap}_{ijk} = \mu + \mathbf{x}_{ijk}^\top \boldsymbol{\beta} + u_k + v_{jk} + \varepsilon_{ijk}, \quad (2)$$

where $\mathbf{x}_{ijk}$ are standardized question-level covariates, $u_k \sim \mathcal{N}(0, \sigma_{\text{domain}}^2)$ is a domain random intercept, $v_{jk} \sim \mathcal{N}(0, \sigma_{\text{subject}}^2)$ is a subject-within-domain random intercept, and $\varepsilon_{ijk} \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ is residual noise. The model is estimated by restricted maximum likelihood (REML) using `statsmodels.MixedLM` [Raudenbush and Bryk, 2002].

The fixed effects $\boldsymbol{\beta}$ characterize which question features are associated with miscalibration across the full dataset, interpreted as descriptive associations rather than causal effects. The variance components $(\sigma_{\text{domain}}^2, \sigma_{\text{subject}}^2, \sigma_\varepsilon^2)$ answer RQ1 via an intraclass correlation (ICC) decomposition: the fraction of total calibration heterogeneity attributable to each level of the hierarchy. Subject-level shrunk calibration estimates are the fitted random effects $\hat{v}_{jk}$ (BLUPs), which implicitly borrow strength across subjects in proportion to sample size.

4.5 Multiple Testing Correction

For each of the 57 subjects we test H_0 : mean calibration gap = 0 (no systematic miscalibration) against the two-sided alternative using a one-sample t -test on the question-level gaps. We apply the Benjamini–Hochberg procedure [Benjamini and Hochberg, 1995] at $\alpha = 0.05$ to control the false discovery rate, yielding a ranked list of subjects where miscalibration is statistically established rather than just descriptively apparent.

4.6 Cross-Model Replication

The full pipeline is run on each of the four models. Subjects flagged as miscalibrated by multiple models suggest task-driven miscalibration reflecting question structure; subjects that diverge suggest model-specific effects of training. We compare subject-level ECE rankings across all six model pairs using Spearman rank correlation with bootstrap confidence intervals (2000 replicates, seed 42).

5 Results

5.1 Variance Decomposition (RQ1)

Table 1 reports the ICC decomposition from the three-level mixed model for each of the four models. Across all four, the bulk of calibration-gap variance lies at the question level (92.2–98.1%), and domain-level variance collapses to zero. Between-subject ICCs differ by nearly a factor of four: GPT-4o-mini concentrates 7.9% of variance at the subject level, Qwen-7B 4.9%, and Llama and Qwen-1.5B around 2%. Within the Qwen family, ICC roughly triples as scale grows from 1.5B to 7B.

Table 1: Variance decomposition of the calibration gap across four models. ICC columns are the share of total variance attributable to subject-level heterogeneity. Domain-level variance is estimated at zero for all four models.

Model	σ_{domain}^2	$\sigma_{\text{subject}}^2$	σ_{ε}^2	ICC _{subject}
GPT-4o-mini	0.000	0.013	0.150	0.079
Llama-3-8B-Instruct	0.000	0.004	0.190	0.020
Qwen-2.5-1.5B-Instruct	0.000	0.004	0.203	0.019
Qwen-2.5-7B-Instruct	0.000	0.009	0.177	0.049

The domain-level variance is estimated at zero for every model. With only four domain groups, the likelihood is flat with respect to the domain variance component, and the estimate collapses to the boundary. We therefore report total between-subject variance without a clean domain/subject partition, and treat domain as a descriptive grouping rather than a modeled random effect.

5.2 Calibration Curves by Subject

Figure 1 shows reliability curves for the five most miscalibrated subjects by ECE for GPT-4o-mini. The curves are masked outside the observed 1st–99th percentile of confidence per subject to avoid displaying isotonic extrapolation: GPT-4o-mini’s confidence is so concentrated above 0.7 that any plotted curve below this region would be pure extrapolation. Within the observed support, all five subjects show the characteristic shape of an overconfident model—empirical accuracy lying below the diagonal, sometimes sharply.

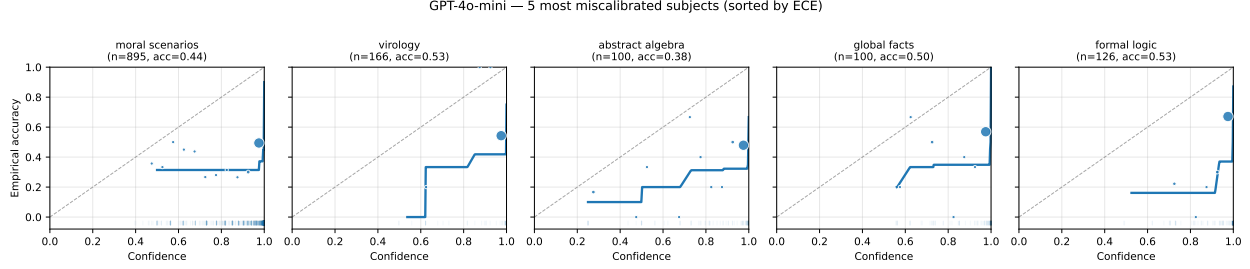


Figure 1: GPT-4o-mini reliability curves for the five most miscalibrated subjects by ECE. Solid line: isotonic regression fit, plotted only over the observed confidence range (1st–99th percentile per subject). Markers: empirical accuracy in 20 equal-width confidence bins, sized by bin count. The dashed line is the calibration diagonal. Annotations report sample size and mean accuracy.

For visual contrast, Figure 2 shows the same plot for the five *best*-calibrated subjects for GPT-4o-mini. Curves there track the diagonal closely or lie just below it; both shapes are present in the same model on the same evaluation, illustrating the heterogeneity that aggregate ECE conceals.

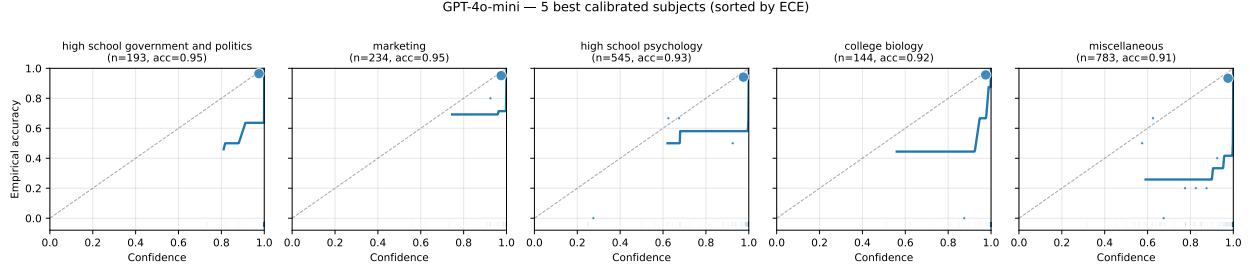


Figure 2: GPT-4o-mini reliability curves for the five *best*-calibrated subjects, shown for visual contrast with Figure 1. Same conventions.

5.3 Predictors of Miscalibration (RQ3)

Table 2 reports the fixed effects from the multilevel model across all four models. The intercepts confirm large mean overconfidence in every model, ranging from 0.157 (Qwen-1.5B) to 0.233 (Qwen-7B). Within the Qwen family, the larger 7B model is *more* overconfident than the smaller 1.5B on average (+49% in the intercept), despite higher accuracy (0.685 vs 0.536)—consistent with the documented effect of RLHF-style alignment applied more aggressively to stronger base models [Kadavath et al., 2022, Ouyang et al., 2022, Nakkiran et al., 2025].

Figure 3 shows the corresponding forest plot. Word count is a positive significant predictor for the two larger models (GPT-4o-mini, Qwen-7B) and inert for the smaller two (Llama, Qwen-1.5B). The most informative pattern is the entropy coefficient, which is positive for GPT-4o-mini (+0.016, $p < 0.001$) and Qwen-7B (+0.029, $p < 0.001$), marginal for Llama (+0.008, $p = 0.07$), and negative for Qwen-1.5B (−0.010, $p = 0.02$).

The entropy result is initially counterintuitive: higher option-distribution entropy mechanically reduces top-1 confidence (the maximum probability over a flatter distribution must be lower), so a naïve prediction is that high-entropy questions should show *smaller* calibration gaps. Figure 4 traces what is happening. Within each model, we partition questions into entropy quartiles and report mean confidence (solid line) and mean accuracy (dashed line) within each. As entropy rises,

Table 2: Fixed effects from the three-level mixed model for all four models. Continuous covariates standardized. Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$.

Predictor	GPT-4o-mini	Llama-Instruct	Qwen-1.5B-Instruct	Qwen-7B-Instruct
Intercept	0.196***	0.197***	0.157***	0.233***
Word count	0.031***	-0.005	0.004	0.019**
Max option length	0.001	-0.001	-0.002	-0.007
Negation	-0.001	0.007	-0.010	-0.001
Entropy	0.016***	0.008†	-0.010*	0.029***

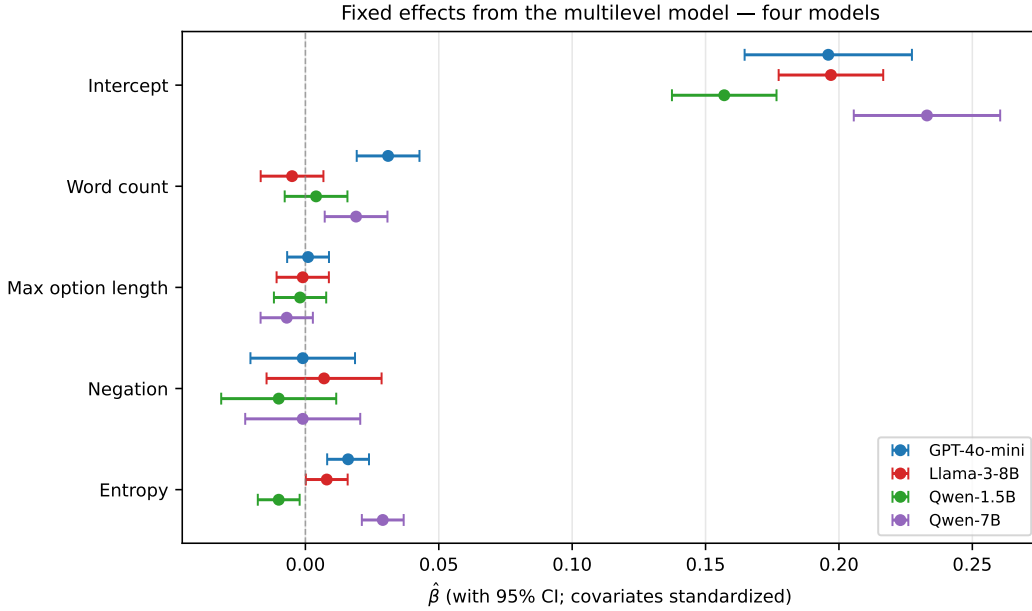


Figure 3: Fixed effect estimates with 95% confidence intervals for all four models. Estimates are on the standardized calibration gap scale.

confidence does fall in all four models, but in three of them accuracy falls *faster*, so the gap (shaded area) widens. Qwen-1.5B is the lone exception: its confidence and accuracy fall in roughly parallel across the entropy range, which is exactly why its entropy coefficient is negative.

This marginal pattern is consistent with a literature on RLHF-induced confidence saturation [Kadavath et al., 2022, Nakkiran et al., 2025, Yaldiz et al., 2026], which finds at the aggregate level that instruction-tuned models maintain high top-1 probability even when their option distribution suggests genuine uncertainty. We are cautious about extending this further: the negative-coefficient result rests on a single small-model checkpoint (Qwen-1.5B), and the size differences between coefficients reflect observational rather than controlled variation. Pending replication with additional small or base models, our claim is descriptive: the four models we examine exhibit a sign flip in the entropy coefficient that aligns with the direction prior literature predicts.

5.4 FDR-Corrected Subject Rankings

Figure 5 shows all 57 subjects sorted by mean calibration gap for each model, with FDR-rejected subjects highlighted. At $\alpha = 0.05$, all 57 subjects are rejected for GPT-4o-mini, Llama, and

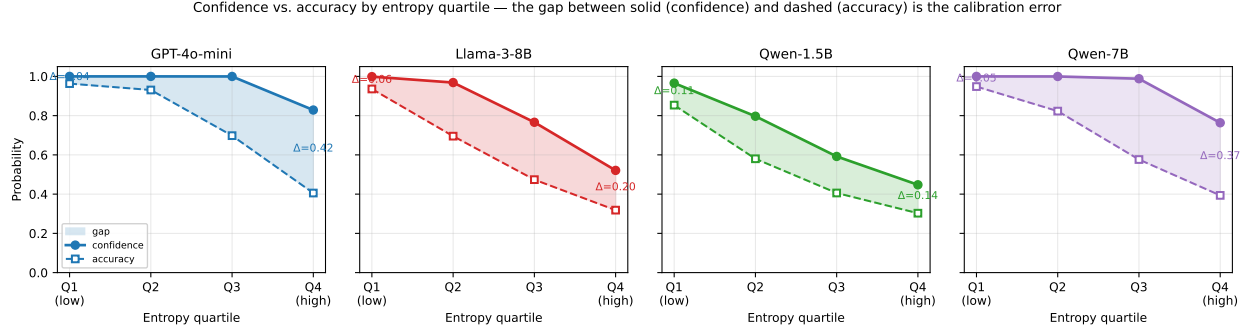


Figure 4: Confidence (solid) and accuracy (dashed) by entropy quartile, per model. The shaded area is the calibration gap. Three of four models show confidence falling more slowly than accuracy, widening the gap; only Qwen-1.5B shows confidence and accuracy falling in parallel.

Qwen-7B; for Qwen-1.5B, 54 of 57 are rejected. The three subjects that fail to reject for Qwen-1.5B are `high_school_government_and_politics` ($\bar{g}_{\text{ap}} = 0.025$, $p_{\text{adj}} = 0.37$), `sociology` ($\bar{g}_{\text{ap}} = 0.033$, $p_{\text{adj}} = 0.24$), and `us_foreign_policy` ($\bar{g}_{\text{ap}} = 0.057$, $p_{\text{adj}} = 0.14$). These are the only subject-by-model cells in the entire 4-model $\times$ 57-subject study where calibration is statistically indistinguishable from zero.

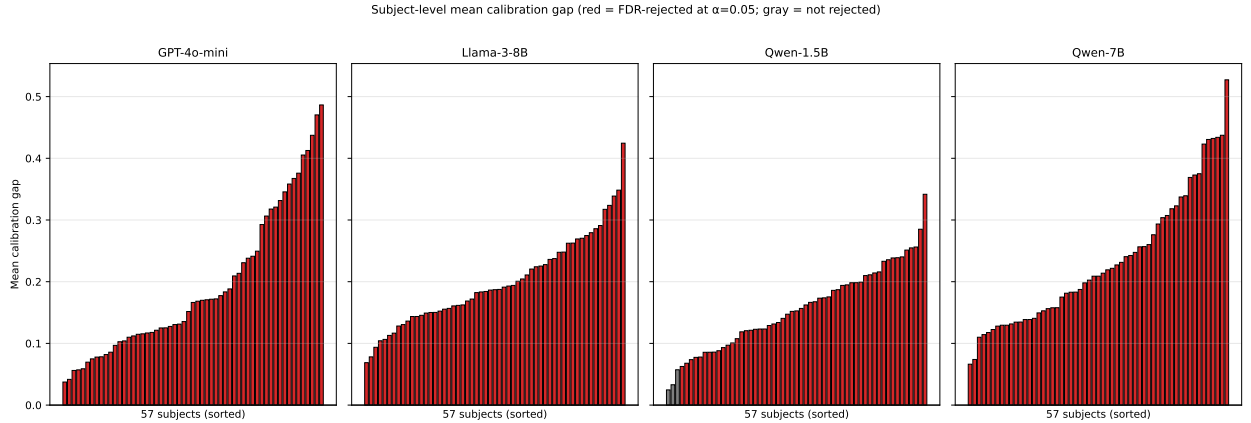


Figure 5: Mean calibration gap per subject, sorted ascending. Red bars indicate subjects where H_0 : mean gap = 0 is rejected at FDR $\alpha = 0.05$; gray bars are not rejected. Three of four models reject all 57 subjects; Qwen-1.5B rejects 54 of 57.

The substantive finding is therefore the *magnitude* of heterogeneity, not the binary reject/fail-to-reject outcome. Mean gaps span a $13.0\times$ range for GPT-4o-mini (0.037 to 0.487), $6.2\times$ for Llama (0.069 to 0.425), $13.9\times$ for Qwen-1.5B (0.025 to 0.342), and $8.0\times$ for Qwen-7B (0.066 to 0.527).

Subjects that are robustly worst-calibrated. Across the four models, the same subject clusters appear at the high-ECE end of the distribution. Four subjects appear in the top-10 most miscalibrated for all four models: `virology`, `college_chemistry`, `college_physics`, `professional_law`. Expanding to top-15 adds `college_mathematics`, `econometrics`, `high_school_physics`, `machine_learning`, and `moral_scenarios`. These naturally cluster into three semantic groups: (i) college-level quantitative STEM, (ii) niche factual recall (virology, global facts), and (iii) structured ethical/legal

reasoning (moral scenarios, formal logic, professional law).

Subjects that are robustly best-calibrated. Symmetric to the above, the same subjects appear at the low-ECE end. The intersection of bottom-5 ECE across all four models contains `high_school_government_and_politics` and `high_school_psychology`; bottom-10 adds `high_school_us_history`, `marketing`, and `miscellaneous`; bottom-15 adds `high_school_geography`, `high_school_world_history`, `sociology`, and `us_foreign_policy`. Two patterns are visible. First, MMLU’s high-school-level subjects systematically show smaller calibration gaps than their college-level counterparts; the level of specialization required appears to track miscalibration. Second, well-calibrated subjects are dominated by Social Sciences and the “Other” category (the applied/factual subjects in the MMLU taxonomy)—no STEM subject appears in the bottom-10 intersection. The audit recovers a coherent contrast: applied/factual high-school content is where calibration is reliably good, and college-level technical content is where it is reliably bad.

5.5 Sensitivity Analysis

Sensitivity-analysis conclusions extend uniformly across all four models. ECE rankings are stable across bin counts (Spearman ≥ 0.97 vs. the 20-bin baseline for every model). The Negative Log Calibration Score, a strictly proper scoring rule, ranks subjects nearly identically (Spearman ≥ 0.85 across models and bin counts). Two-level and three-level ICC estimates are numerically identical, consistent with σ_{domain}^2 collapsing to zero.

The FDR-rejection finding is also robust to threshold choice for the three larger models. At $\alpha = 0.01$, GPT-4o-mini rejects 56/57, Llama 55/57, and Qwen-7B 57/57. Qwen-1.5B is more sensitive: at $\alpha = 0.01$, only 46 of 57 are rejected. This is consistent with Qwen-1.5B’s lower mean intercept (0.157) and broader distribution of subject-level p -values—the smallest model in our set is the only one where many subjects’ miscalibration is detectable but not robust under the strictest FDR threshold.

5.6 Cross-Model Agreement (RQ2)

Figure 6 shows the 4×4 Spearman rank correlation matrix for subject-level ECE across the four models. All six pairwise correlations fall between 0.78 and 0.92. The cross-family pair GPT-4o-mini vs Qwen-7B is the highest at 0.924, exceeding even the within-Qwen scaling correlation (1.5B vs 7B, 0.849). Cross-family agreement at matched capability is therefore stronger than within-family agreement across scale—the cleanest evidence in our data that subject-level miscalibration is primarily a property of the *task* rather than of any particular model’s training.

Bootstrap 95% confidence intervals (2000 replicates) for each pairwise correlation are: GPT vs Llama [0.73, 0.91], GPT vs Qwen-1.5B [0.64, 0.87], GPT vs Qwen-7B [0.85, 0.96], Llama vs Qwen-1.5B [0.73, 0.89], Llama vs Qwen-7B [0.80, 0.93], Qwen-1.5B vs Qwen-7B [0.76, 0.90]. None of these intervals contains zero or even values close to it, and the lowest lower bound (0.64) is still substantively large.

6 Discussion

Where calibration succeeds versus fails. The two strongest findings are descriptive but coherent. Calibration is reliably good on high-school-level Social Sciences and applied/factual subjects (`marketing`, `miscellaneous`, `high_school_government_and_politics`, `high_school_psychology`), and reliably bad on college-level technical content (`college_chemistry`, `college_physics`, `college_mathematics`,

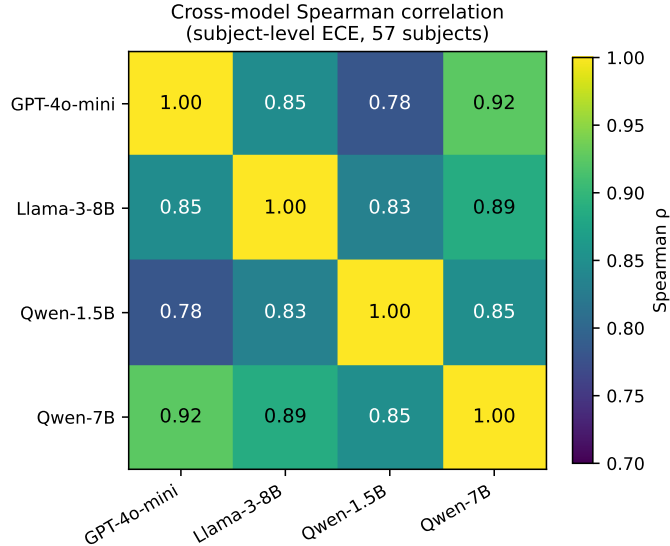


Figure 6: Spearman rank correlation of subject-level ECE across the four models (57 common subjects). All pairwise correlations are in $[0.78, 0.92]$; bootstrap 95% CIs are reported in the text.

`machine_learning`), niche factual recall (`virology`, `global_facts`), and structured ethical/legal reasoning (`moral_scenarios`, `professional_law`, `formal_logic`). The same patterns appear in every model we examined. The level of specialization required appears to be a meaningful axis: MMLU’s high-school-level subjects consistently show smaller calibration gaps than their college-level counterparts. This is the substantive finding the heterogeneity audit recovers, and the kind of structure aggregate ECE conceals.

Cross-family agreement at matched capability. The Spearman correlation between GPT-4o-mini and Qwen-7B (0.924) exceeds the within-family Qwen scaling correlation (0.849). All six pairwise correlations fall in $[0.78, 0.92]$. Subject-level miscalibration is therefore more consistent across model families at matched capability than within a single family across scale—strong evidence that miscalibration patterns are primarily a property of the task, not the model. This stability provides the structural justification for choosing to scope corrections at the subject level rather than globally; the targets of any such correction are stable across the model space we examined, even though each correction would still be fit per model on held-out data.

Within-family scale and entropy: descriptive observations. The Qwen 1.5B $\rightarrow$ 7B comparison shows the intercept rising from 0.157 to 0.233 (+49%), ICC roughly tripling (0.019 $\rightarrow$ 0.049), and the entropy coefficient flipping sign ($-0.010 \rightarrow +0.029$). All three changes go in the same direction: more capable Qwen $\rightarrow$ more concentrated overconfidence. We present this as an observation about the four checkpoints we measured rather than a controlled scaling experiment. Figure 4 traces the entropy effect to a marginal pattern in which capable models maintain near-saturated confidence even as the option distribution flattens, which is consistent with prior literature on RLHF-induced confidence saturation [Kadavath et al., 2022, Nakkiran et al., 2025, Yaldiz et al., 2026]. We do not claim to have identified a causal mechanism; the comparison is between checkpoints rather than a controlled within-checkpoint manipulation.

Implications for recalibration. Our framework is the diagnostic arm of a broader recalibration picture. The operational arm has been studied directly: Luo et al. [2025] fits subject-specific temperature parameters per MMLU subject and demonstrates improvements over a global correction. Our contribution is the upstream question: is subject-level scoping a structurally appropriate choice in the first place? The audit’s affirmative answer—stable, cross-model targets with robust statistical support—supplies that justification. Practitioners can also reuse the audit findings as a transferable diagnostic test set: the robustly worst-calibrated subjects we identify can flag miscalibration on a new model before deployment, without re-running the full audit.

Limitations. Several limitations warrant caution. First, domain-level variance is estimated at zero, likely because subject-level random effects absorb whatever between-domain structure exists; with only four domain groups, power to detect a separate domain component is very low, so we cannot cleanly partition between-subject variance into domain-level and subject-within-domain components. Second, the mixed model assumes homoskedastic residuals, which is violated when calibration gap variance depends on per-subject accuracy. Third, confidence is computed from logprobs at a single token position; this may not faithfully represent the model’s internal uncertainty, especially after RLHF fine-tuning [Kadavath et al., 2022]. Fourth, results are specific to MMLU’s four-choice format and may not generalize to open-ended generation settings. Fifth, 4-bit quantization affects three of the four models studied and may modestly attenuate confidence—most notably for Qwen-1.5B, whose test-set accuracy (0.536) is below its published full-precision figure; re-running this model under different precision settings would clarify how much of its anomalous behavior (negative entropy coefficient, FDR robustness gap) survives at full precision. Sixth, the entropy-coefficient sign comparison rests on a single small-model checkpoint (Qwen-1.5B); replication with additional small or base models—particularly within the Llama family—would test whether the pattern is monotonic with capability or specific to that checkpoint.

Future directions. Three directions extend the framework most directly. First, additional scale points within the Qwen family (e.g., Qwen-2.5-14B) would let us fit a continuous scaling relationship for the intercept and entropy coefficient rather than contrasting two checkpoints. Second, adding a third model family (e.g., Mistral-7B or Gemma-2-9B at matched scale) would strengthen the cross-family generalization claim from two families to three. Third, applying the framework to MMLU-Pro or other harder benchmarks would test whether the high-school-vs-college calibration gap we observe within MMLU persists at higher difficulty levels.

7 Conclusion

We presented a statistical audit framework for LLM calibration heterogeneity and applied it to MMLU using four instruction-tuned models from two families across three parameter scales. The framework recovers a coherent contrast in where calibration succeeds versus fails: applied/factual high-school subjects are reliably well-calibrated, while college-level technical content, niche factual recall, and structured ethical/legal reasoning are reliably bad. Subject-level ECE rankings are highly correlated across all six model pairs (Spearman $\rho \in [0.78, 0.92]$), with the cross-family GPT-4o-mini vs Qwen-7B correlation being the highest. A within-family scale comparison (Qwen-1.5B vs Qwen-7B) shows the intercept, between-subject ICC, and entropy coefficient all changing in the direction prior aggregate literature predicts—an observational pattern consistent with RLHF-induced confidence saturation rather than a controlled mechanism test. The audit’s stable targets provide the structural justification for choosing subject-aware over global recalibration scoping—a choice

operationalized in concurrent work by Luo et al. [2025]—and supply a transferable diagnostic for flagging miscalibration on new models before deployment.

Statement of Work

Data analysis. Jennifer Zhang conducted the original two-model analysis (GPT-4o-mini, Llama-3-8B-Instruct), including the multilevel model, calibration curves, FDR correction, and sensitivity checks. John Wright collected and analyzed the two additional Qwen-2.5 models, extended the multilevel model and cross-model framework to four models, and produced the within-family scale and entropy-mechanism analyses. **Coding.** Jennifer Zhang built the original pipeline; John Wright extended it to support arbitrary models and produced the parameterized inference notebook used for the open-weight models. **Writing.** Jennifer Zhang drafted the original report; John Wright revised it for the four-model expansion and produced the final manuscript. **Presentation.** Jennifer Zhang prepared the original slide deck.

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